

RoadSafe Analytics Project Documentation

Methodology

The **RoadSafe Analytics** project follows a structured data analytics workflow to ensure accuracy, scalability, and meaningful insight generation from large-scale road accident data (2016–2023).

1. Data Collection & Integration

- Accident datasets were sourced from publicly available US road accident records.
- Multiple CSV files were merged sequentially to maintain temporal continuity.
- Data was loaded into a PostgreSQL database for efficient querying and filtering.

2. Data Cleaning & Preprocessing

- Handled missing values using appropriate strategies (imputation or removal based on impact).
- Standardized categorical variables such as state, weather condition, time of day, and severity.
- Converted date-time fields into proper formats to enable time-based analysis.
- Removed duplicates and corrected inconsistent entries.

3. Feature Engineering

- Derived new features such as accident year, month, weekday, and visibility ranges.
- Grouped continuous variables (e.g., visibility, temperature) into meaningful bins.
- Created severity and risk-based categorizations for better interpretation.

4. Exploratory Data Analysis (EDA)

- Used statistical summaries and visualizations to identify trends and patterns.
- Analyzed accidents by state, time of day, weather conditions, and severity.
- Identified high-risk regions and peak accident periods.

5. Visualization & Dashboard Development

- Built interactive visualizations using Matplotlib and Seaborn.
- Developed a Streamlit-based dashboard for real-time filtering and exploration.
- Enabled multi-parameter filtering (state, weather, severity, time range).

Key Insights

1. State-wise Analysis

- California consistently report the highest number of accidents, indicating a strong correlation with population density and traffic volume.

2. Time-Based Trends

- Accident frequency is significantly higher during evening and night hours, suggesting reduced visibility and driver fatigue as major contributing factors.

3. **Weather Conditions**

- Cloudy and rainy conditions show a noticeable increase in accident severity compared to clear weather.

4. **Severity Distribution**

- Moderate severity accidents dominate the dataset, but high-severity cases increase during poor visibility conditions.

5. **Visibility Impact**

- Accidents sharply increase when visibility falls below a critical threshold, highlighting the importance of road lighting and weather alerts.

6. **Yearly Trends**

- A gradual increase in reported accidents is observed post-2020, possibly due to improved reporting systems and increased vehicle usage.
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Conclusions

- Road accidents are not random events; they are strongly influenced by **time, location, weather, and visibility**.
 - High-accident states require targeted policy interventions such as improved traffic management and infrastructure upgrades.
 - Night-time driving and adverse weather conditions pose the highest risk, emphasizing the need for better lighting, warning systems, and driver awareness programs.
 - Data-driven dashboards like RoadSafe Analytics can support **government agencies, traffic planners, and law enforcement** in making informed decisions.
 - The project demonstrates how combining **EDA, visualization, and interactive analytics** can transform raw accident data into actionable road safety insights.
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Overall Outcome:

RoadSafe Analytics successfully converts large-scale accident data into meaningful insights that can help reduce accidents, improve road safety policies, and save lives.